Homolog-expanded enzyme adaptation of a from-scratch protein language model (LIKHA): per-family gains on lignocellulose enzymes

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Abstract LIKHA is a 354M from-scratch protein language model whose first enzyme-specialization adapter, trained on 555 reviewed lignocellulose-enzyme sequences, gave only a small perplexity gain and a family-tag signal that behaved as a generic "enzyme-like" switch rather than family-specific control. We ask whether expanding the training data with evolutionary homologs helps. We pull UniRef90 homologs of four rice-straw lignocellulose enzyme families (3,392 sequences after a benign floor and dedupe, vs 657 reviewed before), build a cluster-disjoint, exact-identity leak-checked split (2,817 train / 419 validation used at a 1,000-residue cap), and train a tag-aware LoRA adapter (rank 8, general-protein replay, only four new tag-embedding rows trainable). The adapter lowers held-out enzyme NLL with no general-protein forgetting (general-val NLL change −0.0008; base embeddings verified unchanged), and the gains concentrate on the hardest, most data-poor families: endo-1,4-beta-xylanase perplexity falls 10.6 to 8.1 (about 24%), while the already well-modelled beta-glucosidase moves only 4.7 to 4.5 (about 4%). The pooled metric understates this because it is dominated by the easy family. Homolog expansion is the right lever where it matters; the undertrained 354M base remains the ceiling on the size of the gain. pLDDT and perplexity here are computational, with no wet-lab validation.

Status: experiment finished, results fetched and sha-verified. INTERNAL, no release, no served endpoint. Base weights never changed.

1 Introduction

LIKHA stage 1 is an internal, from-scratch protein language model (354,280,448 parameters, char-level amino-acid tokenizer, domain tags). A prior adapter specialized it toward rice-straw lignocellulose enzymes (endoglucanase EC 3.2.1.4, cellobiohydrolase 3.2.1.91, beta-glucosidase 3.2.1.21, endo-xylanase 3.2.1.8) using 555 reviewed Swiss-Prot sequences; the held-out enzyme perplexity gain was about 10% on a 102-sequence family validation set, and a counterfactual analysis showed the per-family tags acted as a broad "enzyme-like" switch, not family-specific control. The limiting factor looked like data: a few hundred reviewed sequences per family.

This report tests the fix. INTERNAL, no release.

Contributions:

1. **A homolog-expanded, leak-safe enzyme dataset** — 3,392 benign-floored UniRef90-homolog sequences (6x the reviewed set), split cluster-disjoint with an exact-identity (>=0.8) leak check.
2. **A measured per-family result** — the adapter lowers held-out enzyme NLL most on the hardest families (xylanase ~24% perplexity reduction) with no general-protein forgetting, and the shipped adapter changes no base weight.
3. **A caution about the pooled metric** — pooled NLL hides the gains because one easy family dominates the validation mix; per-family reporting is required.

2 Method

Data. UniRef90 homologs of the four seed families were pulled (data-engineer, scope B) with the standard benign floor (exclude KW-0800 toxin, KW-0843 virulence, select-agent lineage): 3,392 distinct sequences kept (624 reviewed), 528 clusters, 2,684 Eukaryota / 708 Bacteria. The pull was sha-verified (family_all.jsonl sha 3778d869). A cluster-disjoint split was produced with the audited MMseqs2 harness (min-seq-id 0.8/0.9), with an exact-identity leak search that dropped 106 validation sequences at >=0.8 identity to train and two live controls (exact-copy identity 1.0, 10%-mutant 0.902, both caught). The split: 2,879 train / 407 validation; at the 1,000-residue cap the adapter used 2,817 train / 419 validation sequences.

Adapter. Tag-aware LoRA on the frozen bf16 base: rank 8, lr 1e-4, four new EC tag tokens in unused vocabulary ids 34-37 with their own learning rate (1e-3) and only those four tied-embedding rows trainable (gradient mask plus a zero-weight-decay group). General-protein replay (beta 0.3) against a held-out general set guards against forgetting. 800 steps, batch 8, one H100 SXM (SECURE), bf16; the base weights are never updated. Evaluation every 100 steps reports pooled and per-EC tagged validation NLL and the general-protein validation NLL.

3 Experiments

Setup. Metric: mean per-residue negative log-likelihood (NLL) on the cluster-disjoint enzyme validation set, tagged (the EC tag prepended) and pooled across the four families, plus the per-EC breakdown; and the general-protein validation NLL as a forgetting control. Baseline is the frozen base with the tags but no adapter.

Result 1: per-family gains, concentrated on the hard families. Best-checkpoint perplexity (exp of NLL):

Enzyme family	EC	baseline ppl	best ppl	reduction	best step
Endo-1,4-beta-xylanase	3.2.1.8	10.61	8.07	~24%	800
Endoglucanase (GH5/9)	3.2.1.4	6.51	6.07	~7%	700
Cellobiohydrolase	3.2.1.91	9.20	8.59	~7%	800
beta-glucosidase	3.2.1.21	4.70	4.49	~4%	200

The three hardest, most data-poor families keep improving to steps 700-800; only the already well-modelled beta-glucosidase peaks early (step 200) and then overfits. So 800 steps is appropriate for the hard families; the pooled curve's apparent overfitting after step 200 is the easy family's overfit, weighted heavily in the pool.

Result 2: pooled understates the gain. Pooled tagged validation NLL moves 1.836 (baseline) to a best of 1.7993 (step 200) and 1.8176 (final), a perplexity change of only 6.27 to 6.05 (~3.6%). The pool is dominated

by the 3.2.1.21 validation sequences (183 of 419), the family with the least headroom, which masks the large xylanase gain.

Result 3: no general-protein forgetting. General-protein validation NLL is 1.9474 (baseline) and 1.9466 (final), a change of −0.0008. The replay term holds the base distribution. The adapter's gradient mask is verified: base_embeddings_unchanged = true (only the four tag rows moved).

Cost: USD 0.662, 10.0 minutes on one H100 SXM; pod self-terminated. Adapter adapter_model.safetensors sha 6087ec8c (5.53 MB); results.tgz sha a2f1d9d0.

4 Discussion and Limitations

The homolog expansion does what the reviewed-only set could not: it lifts the families that were starved of data. Xylanase, the hardest family at baseline, gains the most (~24% perplexity). This supports the earlier conclusion that homolog expansion, not family tags, is the real lever for enzyme specialization.

The gains are real but modest in absolute terms (fractions of a nat). Two limits bound them. First, the base: a 354M from-scratch model whose stage-1 schedule was cut at 87% is the ceiling on how much any adapter can add. Second, the metric: NLL and any downstream pLDDT are computational proxies, with no wet-lab validation of a single generated or scored sequence.

This is not a like-for-like comparison with the 555-sequence reviewed adapter: the validation sets differ (419 homolog-split sequences here vs 102 reviewed-family sequences there), so the pooled perplexity numbers are not directly comparable; the per-family direction is the robust reading. The leak check de-leaks within the homolog set (>=0.8 identity) but was not run against the earlier reviewed validation set. A single training run, one seed, no confidence intervals on the NLL deltas.

5 Conclusion

Homolog-expanded data strengthens LIKHA's modelling of rice-straw lignocellulose enzymes, most on the hardest, data-poor families (xylanase ~24% perplexity reduction), with no general-protein forgetting and no change to the base weights. The single most useful next step is a shorter-schedule, per-family-early-stopped adapter on the same data, and, for a larger gain, a better-trained base.

6 Acknowledgments

devops lane (execution; research and data-science held by devops while those lanes are off biology). Data pull and cluster-disjoint split by the data-engineer lane (benign floor, audited MMseqs2 harness). Base model: LIKHA stage 1 (internal). Compute: one H100 SXM (SECURE), self-terminating. AI assistance: analysis and drafting with Claude (Anthropic); every number is read from the run's lora_results.json.

7 References

Internal: docs/LIKHA-STAGE1-REPORT-2026-11-06.md; docs/LIKHA-AGRIWASTE-ENZYME-PLAN-2026-11-07.md; experiments/likha-enzyme-lora-2026-11-07/ (the earlier 555-sequence adapter and counterfactual). Data receipts: D:-homolog-v0_1.json, likha-homolog-split-v0_1.json.